

Home

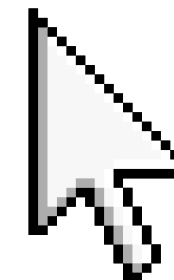
Content

Contact

Introduction to

DEFENSE IN DEPTH

A LAYERED SECURITY STRATEGY



Start



Picture Content

Contact

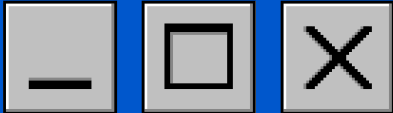
Policy Management & Prevention

- Continuous C&A
- IT Security Governance
- Security Plan
- Vulnerability Assessment
- Business Continuity Plan
- Security Awareness Training
- IT Disaster Recovery Plan
- Security Architecture & Design
- Risk Management
- Security Policies & Compliance
- Data Classification



Monitoring & Response / Operations

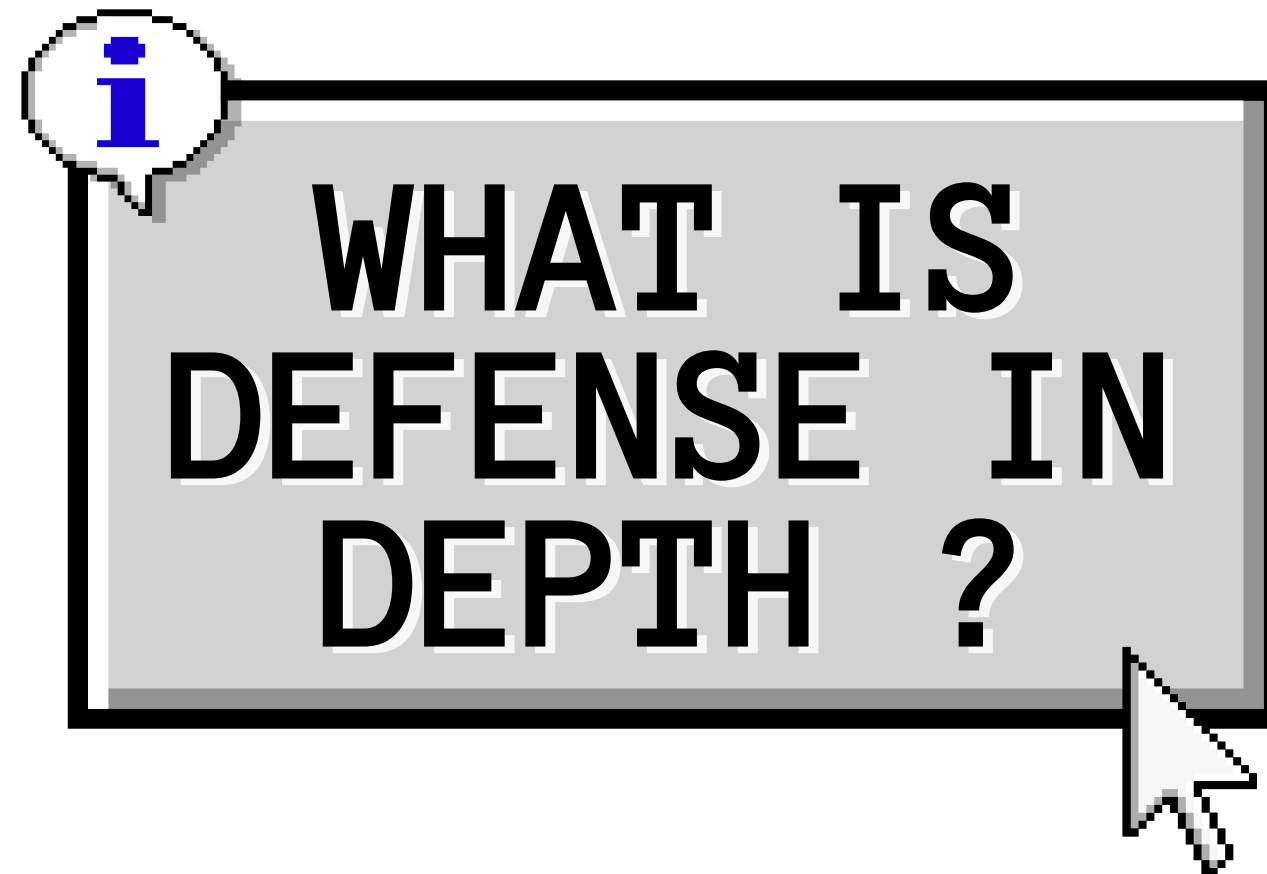
- CSIRT
- SIEM
- Focused Ops
- SIEM
- Security Dashboard
- Digital Forensics
- SOC/NOC Monitoring (24x7)
- Escalation Management
- Continuous Monitoring & Assessment Situational Awareness
- Security SLA/SLO Reporting
- Penetration Testing
- Cyber Threat Intelligence



Picture Content

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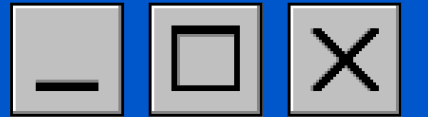
Perimeter Security	Network Security	Endpoint Security	Application Security	Data Security
• ICAM/ZTA (2FA/MFA) • DDos Prevention • Message Security • Firewall • Web Content Filter • IDS/IPS • DMZ • VPN • 2FA/MFA • WAF • Honeypot • External Vuln Scanning • DLP • Physical Security • Logging	• ICAM/ZTA (2FA/MFA) • Network Device Hardening • Internal Firewall • Internal IDS/IPS • Virtual Patching • VoIP Security • Wireless Security • Mobile Security • NAC • Internal Vuln Scanning • DLP • Physical Security • Logging	• ICAM/ZTA (2FA/MFA) • Endpoint Hardening • Endpoint Firewall • Endpoint White/Blacklist • NGAV/EDR/MDR (malware/ransomware) • File Integrity Monitoring • Patch Management • Endpoint Disk Encryption • DLP • Physical Security • Logging	• ICAM/ZTA (2FA/MFA) • WAF • Database Security • Application Vuln Scanning • Dynamic App Testing • Static App Testing • DLP • Logging	• ICAM/ZTA (2FA/MFA) • Data Classification • Data Integrity Monitoring • Data/Drive Encryption • Data Wiping • PKI • DLP • Logging



- **Definition:** A strategy that uses multiple security measures to protect an organization's assets.
- **Purpose:** If one mechanism fails, another steps up immediately to prevent a breach.

The Layers: Governance → Perimeter → Network → Endpoint → Application → Data → Operations





- **Security Policies & Governance:** The rulebook for how security is handled (People, Process, Tech).
- **Risk Management:** Identifying and prioritizing threats to business operations.
- **Business Continuity & Disaster Recovery (BCP/DR):** Plans to keep the business running during and after a crisis.
- **Security Awareness Training:** Educating employees to be the first line of defense (Human Firewall).
- **Vulnerability Assessment:** Proactive scanning to find weak spots before attackers do.



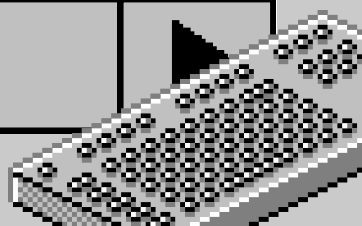
**POLICY,
RISK &
MANAGEMENT**

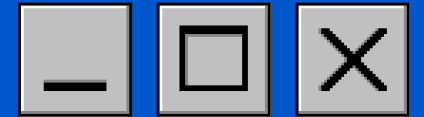




THE OUTER WALL

- **Firewall & WAF:** Filters traffic; blocks unauthorized access to networks and web apps.
- **DDoS Prevention:** mitigates attacks attempting to crash services by flooding them with traffic.
- **DMZ (Demilitarized Zone):** A buffer zone for public-facing services (separates them from the internal LAN).
- **VPN:** Secure, encrypted tunnel for remote access.
- **Honeypot:** A decoy system designed to lure and trap attackers to study their behavior.
- **ICAM / ZTA:** Identity management at the door—verify identity before access.



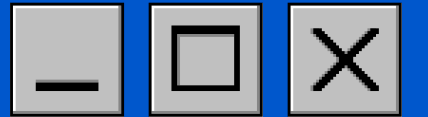


INTERNAL NETWORK DEFENSE

Securing traffic flow inside the organization

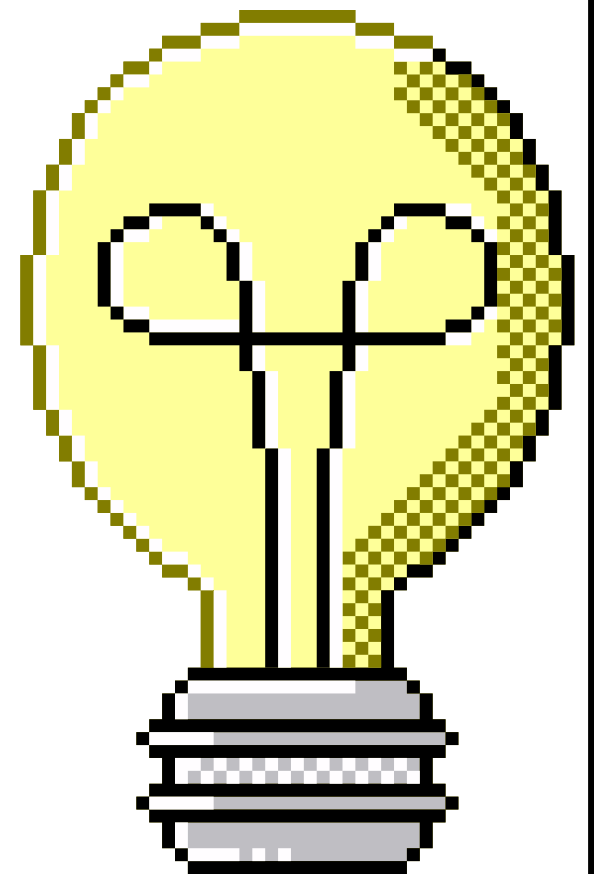
- **Network Access Control (NAC):** Ensures only compliant, authorized devices can connect to the network.
- **Internal Segmentation:** Using internal firewalls/VLANs so attackers cannot move laterally easily.
- **Wireless & Mobile Security:** Securing Wi-Fi (WPA3) and managing mobile devices (MDM).
- **Virtual Patching:** Implementing a security policy to stop an exploit before the vendor releases a real patch.
- **Network Device Hardening:** Removing default passwords and disabling unused ports on switches/routers.

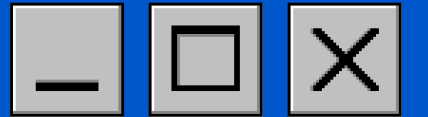




SECURING DEVICES

- **NGAV / EDR / MDR:** Next-Gen Antivirus and Endpoint Detection & Response to catch malware and ransomware.
- **Patch Management:** Keeping OS and software up-to-date to fix known vulnerabilities.
- **Disk Encryption:** Encrypts the hard drive so data is unreadable if the device is stolen.
- **File Integrity Monitoring (FIM):** Alerts if critical system files are changed unauthorizedly.
- **Whitelisting/Blacklisting:** Controlling exactly which applications are allowed to run.





- **SAST & DAST:** Testing code for bugs while writing it (Static) and while it is running (Dynamic).
- **DatabaseSecurity:** Hardening the storage where data lives (Access controls, SQL injection prevention).



- **WAF (Web App Firewall):** Specifically blocks attacks against web apps (like Cross-Site Scripting).
- **Secure Coding:** Designing software with security in mind from day one.

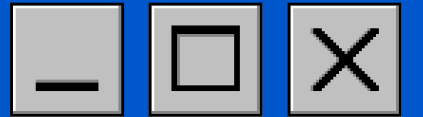


- **Data Classification:** Categorizing data (Public, Internal, Top Secret) to apply the right protection levels.
- **Encryption (At Rest & In Transit):** Making data unreadable to anyone without the key.
- **DLP (Data Loss Prevention):** Detects and blocks sensitive data (like Credit Card #s) from leaving the company.
- **PKI (Public Key Infrastructure):** Manages digital certificates and encryption keys for identity and trust.
- **Data Wiping:** Securely erasing data when it is no longer needed to prevent recovery.



PROTECTING THE CROWN
JEWELS



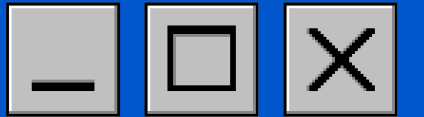


THE WATCHTOWER (OPS)

- **SOC (Security Operations Center):** 24x7 monitoring team.
- **SIEM:** A tool that collects logs from all layers to find patterns of suspicious activity.
- **CSIRT (Incident Response Team):** The "Firefighters" who react when a breach happens.
- **Threat Intelligence:** Knowledge about current hacker tactics used to prepare defenses.
- **Penetration Testing:** Ethical hacking to test if defenses actually work.
- **Digital Forensics:** Investigating evidence after an attack to understand "Who, What, Where."

Cancel





CONCLUSION

